ps0: think about data to answer your questions; produce descriptive stats; due next week jan26

[version: Saturday 31st October, 2026 11:46]

[upto 3 points of extra credit]

1. Getting to know your data takes time: value your time and use data that will serve you beyond this class, say other projects, dissertation, current or future job. And figure out data you'll use in this class asap; at least in the ballpark, can always add other dataset or change dataset later.
2. Think hard about data (and U/A or level of analysis). Write a paragraph or 2 about the data and variables you want to use to answer your research questions.
3. Load the data into Python (as always from online or your own GitHub), and produce very basic descriptive stats of the key variables only, say mean, median, min, and max (and submit ps as per directions below).

HINTS: don't forget about preamble, code comments, u/a, and data url eg: `https://github.com/shirleyrutgers/GIS/blob/main/PS0_FV.ipynb` (yours does not have to be that elaborate in general in preamble at this stage); can just submit Python notebook with writeup in in (in text field), or say link to the writeup elsewhere, say in Google doc.

general directions (always the same):

- i will show your ps in class/repost, as per our core values–opensource and transparency, but if for whatever reason you'd like to keep it private, do let me know, its fine, you just may get less feedback
- you must submit all the code that was executed from the very beginning starting with the very raw data as per replication principle; if data too big to fit online, then just start with eg "to fit data online subset to XYZ regions only" or "took 10% random sample" etc–try to keep the files under 25mb so it loads fast
- ps are cumulative–can and should include much of the previous material; can also use material you've written outside of this class (other classes, projects, etc) but you have to clearly mark the material that has not been written for this class–otherwise, scholastic dishonesty! yes, you can plagiarize yourself!
- use your own dataset; again if you do not have a dataset, ask for help finding it
- you are only submitting code, so it must load data from Internet: `https://theaok.github.io/generic/howToPutDataOnline.html` (when you put data into any public space, try not to violate data copyrights... I haven't had anyone having problems with that, but be careful–for instance you may subset dataset to few vars and smaller sample); and it is also easier to learn on small datasets
- keep it simple! at the beginning of your notebook drop unnecessary vars; and even retain only fewer obs; keep it manageable; much easier to learn using simple data; can always complicate later!; much better to do it right using simple data than do it wrong using complex data
- have nice structure in your file: sections, subsections, comments, etc; may also have multiple files
- great to copy code from others; again, one of the rules for this class is 'be lazy': don't reinvent the wheel, whatever you are coding, has already been done, google things often; but of course you cannot submit 100% code by someone's else, and do cite others incl chatGPT, stack overflow, etc
- if you do something extra/fancy that is relevant, it will be extra credit
- use coding rules that we've learned so far
- submit only the code (Python notebook) and writeup (if separate) into git repo; ps are due by the beginning of the class, ie 6pm; late ps not accepted
- we are on the way to developing the final project with these ps: as we progress, your ps should start resembling a coherent and logical project where you use learned class material to answer interesting questions– say in few sentences (probably at the beginning) why are you doing what you are doing–that is, answer the "so what question": "ok, you're gonna run all that code, and so what?" what's the goal of all that, why are you doing this? you need a compelling justification for what you are doing: say what are those questions you want to answer; related: say why you use data you are using, is it best, does it serve the purpose?
- be prepared do present your code in class (if time), just briefly, key points, couple minutes

- always have exact links to all of the source data (so that i could create the output myself); note: exact links, eg do not say census.gov, but give full url to the data–i must be able o find it; sometimes there is no generic URL–then give steps: what I need to click to get the data!